Ashok Sivathayalan

Ottawa, Ontario

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Education

Carleton University — **Sep. 2021 – April 2026**

Bachelor's of Computer Science Honours (Artificial Intelligence Stream) — *Ottawa, Ontario*

Minor in Biology

11.97/12.00 GPA

Skills

Languages: English (Fluent), French (B2)
Programming Languages: C++, Python, Java, Bash, JavaScript, C, PHP, SQL
Developer Tools: Visual Studio, VS Code, Git, Jira, Docker, Confluence, Figma
Technologies: Linux, Qt, GTest, PyTorch, PostgreSQL, MongoDB, Node.js, IP/TCP

Work Experience

Ford Motor Company — **January 2025 – August 2025**

Software Developer — *Ottawa, Ontario*

- Developed unit tests with C++ and GTest in order to increase coverage by over 1000 lines and hit 90% target on multiple software components
- Created documentation for, led discussion on, and implemented architectural rework to fix fatal bug in API interfacing
- Used C++ and Python to implement new features and fix bugs, and analyze and update functional tests to ensure coverage remains up-to-date

Bombardier Inc. — **May 2024 – December 2024**

Software Developer — *Montréal, Québec*

- Developed application used by simulation team to analyze and compare flight and simulation data
- Wrote Qt C++ Code using Visual Studio IDE to implement and test features

Canadian Bank Note Company — **September 2023 – December 2023**

Software Developer — *Ottawa, Ontario*

- Wrote Bash scripts to aid in CI/CD of auth server for lottery system
- Optimized PHP backend to increase efficiency of online lottery ticket sales process
- Added features to lottery sales website front-end to improve loading times and elevate user experience

Projects

Paper.io Reinforcement Learning | *Python, Gymnasium, RL* — **November 2025**

- Developed custom Gymnasium environment to train RL agents in a Paper.io-based environment
- Used adversarial multi-agent strategy to train agent against itself in the environment

Ontario Bird Classifier | *Python, PyTorch* — **November 2025**

- Created automatic bird classification model with ResNet50 and PyTorch to achieve 82% accuracy through transfer learning and hyperparameter testing
- Enhanced model interpretability by integrating Grad-CAM visualizations to audit neural network decisions and ensure classification based on relevant features

Pac-Man Reinforcement Learning | *Python, Gymnasium, RL* — **November 2025**

- Developed custom Gymnasium environment to test RL agents in a modified Pac-Man environment with stochastic pellet regeneration and high-speed pursuit
- Implemented & conducted systematic hyperparameter testing with SARSA to evaluate agent performance

Relevant Coursework

- Neural Networks
- Generative AI
- Reinforcement Learning
- Software Engineering
- Operating Systems
- Data Structures